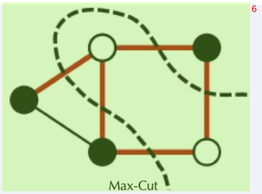
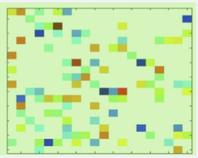
# Fast, Scalable, Warm-Start Semidefinite Programming with Spectral Bundling and Sketching[1]

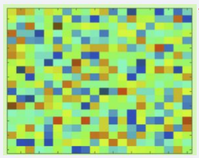
Rico Angell[1] and Andrew McCallum[1][2]
[1]University of Massachusetts Amherst[3]

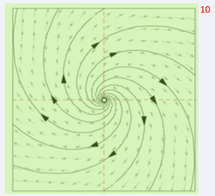
## Motivation[4]

- Semidefinite programming (SDP) is an important tool for many applications[5]

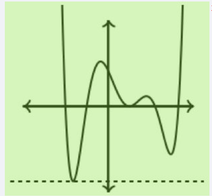

Max-Cut[6]


Matrix Completion[7]

 [9]


Stability of Linear Dynamical Systems[10][12]

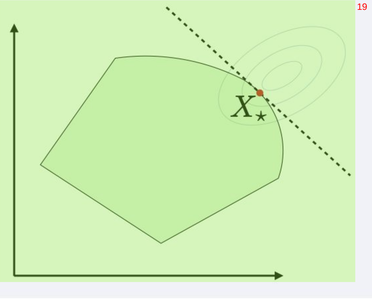
$$(x_1 \vee x_2) \wedge (\neg x_2 \vee x_3)$$
$$(\neg x_1 \vee x_2 \vee \neg x_3)$$
Boolean Satisfiability[13]


Polynomial Optimization[14][15]

- Many existing methods for solving or approximately solving SDPs[16]
- Need for a method that is fast, scalable, and provably correct for massive SDPs
- Need a method where hyperparameters can modulate memory, accuracy, speed, and scalability of solving general SDPs

## Problem Setup[17]

$$\max_{X \succeq 0} \ \langle C, X \rangle$$
$$\text{s.t.} \ \ \mathcal{A}_{\mathcal{I}'} X = b_{\mathcal{I}'}$$
$$\mathcal{A}_{\mathcal{I}} X \leq b_{\mathcal{I}}$$
$\mathcal{A} : \mathbb{S}^n \to \mathbb{R}^m$ linear operator[18]


[19]

- Want to solve general SDPs with both equality and inequality constraints[20]
- Want to have theoretical guarantees about convergence to optimal solution
- Assume cost matrix $C$ is sparse and a upper bound $\alpha \geq \mathrm{tr}(X_\star)$ exists
- Assume the number of constraints is small compared to the problem size
- Want to be able to solve SDPs where $X$ will not fit in memory
- Want solver to be able to take advantage of a warm-start initialization

## Unified Spectral Bundling[21]

- Reformulated SDP as a the following penalized dual objective:[22]

$$f(y) := \alpha\,[\lambda_{\max}(C - \mathcal{A}^* y)]_+ + \langle b, y \rangle + \iota_{\mathcal{Y}}(y)$$
$$= \sup_{(X,\nu) \in \mathcal{X} \times \mathbb{N}} \ \langle C - \mathcal{A}^* y, X \rangle + \langle b - \nu, y \rangle$$[24][25]

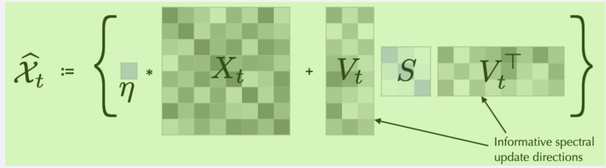
$\iota_{\mathcal{Y}}(y)$: $y_{\mathcal{I}} \geq 0$[23]

$\nu$: Dual slack variable for inequality constraint on $y$[26]

- Solve the SDP by minimizing $f$ using a proximal bundle method:[27]

$$\tilde{y}_{t+1} \leftarrow \arg\min_{y \in \mathbb{R}^m} \hat{f}_t(y) + \frac{\rho}{2}\|y - y_t\|^2$$[28]

where the model $\hat{f}_t$ is parameterized by the low dimensional subspace of $\mathcal{X}$:[29]


[30]

- The size of $V$ determines the rank $k$ of each update, specified by the user[31]
- Only update $y_{t+1} \leftarrow \tilde{y}_{t+1}$ if $\beta(f(y_t) - \hat{f}_t(\tilde{y}_{t+1})) \leq f(y_t) - f(\tilde{y}_{t+1})$
- Always use $\tilde{y}_{t+1}$ to create $\hat{\mathcal{X}}_{t+1}$, $V_{t+1}$ is the top-$k$ eigenvectors of $C - \mathcal{A}^* \tilde{y}_{t+1}$
- Analytical solution to the proximal bundle method lead to the following updates

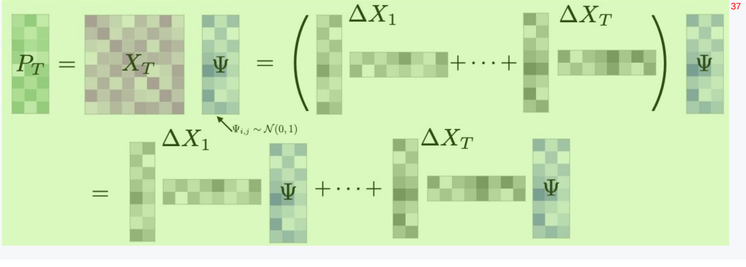
$$\tilde{y}_{t+1} = y_t - \frac{1}{\rho}(b - \nu_{t+1} - \mathcal{A} X_{t+1})$$[32]

$$(X_{t+1}, \nu_{t+1}) \in \arg\max_{(X,\nu) \in \hat{\mathcal{X}}_t \times \mathbb{N}} \ \langle C, X \rangle + \langle b - \nu - \mathcal{A}X, y_t \rangle - \frac{1}{2\rho}\|b - \nu - \mathcal{A}X\|^2$$[33]

- $f(y_t) - \inf f \leq \varepsilon$ after $t \geq O(1/\varepsilon)$ iterations, $O(\cdot)$ suppresses dependence on $n, m$[34]

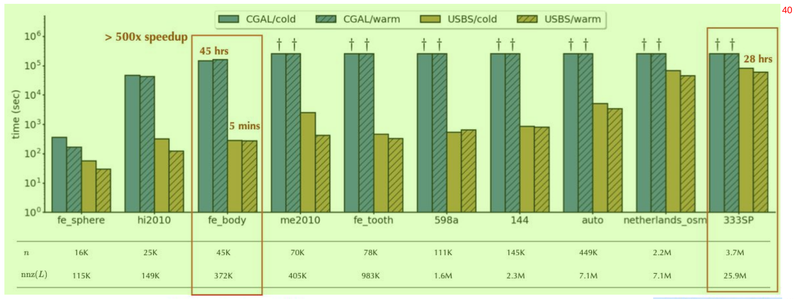
## Scaling with Matrix Sketching[35]

- Only need to keep track of projections of $X$ such as $\langle C, X \rangle$ and $\mathcal{A}X$[36]
- All of the memoization can be done efficiently due to linearity of operations
- Apply low-rank matrix sketching techniques to enable scaling to problems where $X$ cannot fit in memory — track projection $P$ instead of $X$


[37]

## Results[38]

### Max-Cut[39]

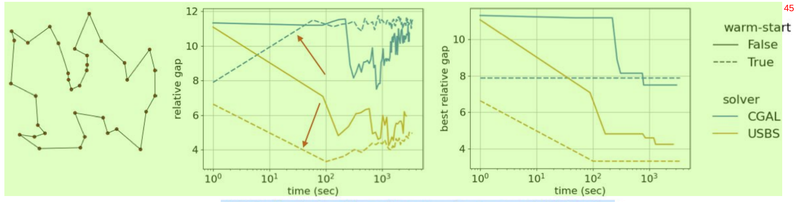

[40]

> 2 billion decision variables[41]

>$10^{13}$ decision variables[42]

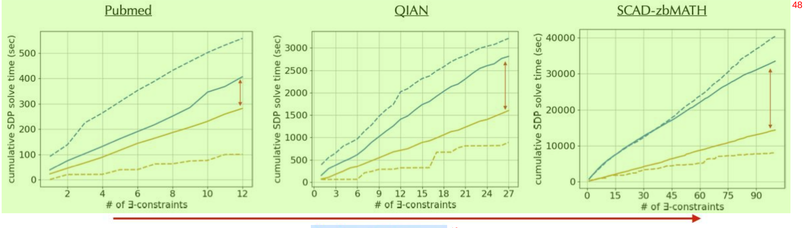
† indicates than an $\varepsilon$-approximate solution was not achieved in 72 hours.[43]

### Traveling Salesman Problem[44]


[45]

USBS effectively taking advantage of warm-start initialization[46]

### Clustering with ∃-constraints[47]


[48]

Problem size increases[49]

UMassAmherst[50]

Manning College of Information & Computer Sciences[51]

[52]

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